10bn Euro space budget
European Space Agency has approved a three-year 10 billion euros budget, although some programmes may be longer or shorter than three years

Moon Phases 2013
Scientific Visualization Studio at NASA has released a new animation showing the phases of the moon in 2013: http://dgit.in/Qhklyd

Lifetimes in space and Earthly homecomings

How will space-faring humans live? Will they ever want to return home to Earth?

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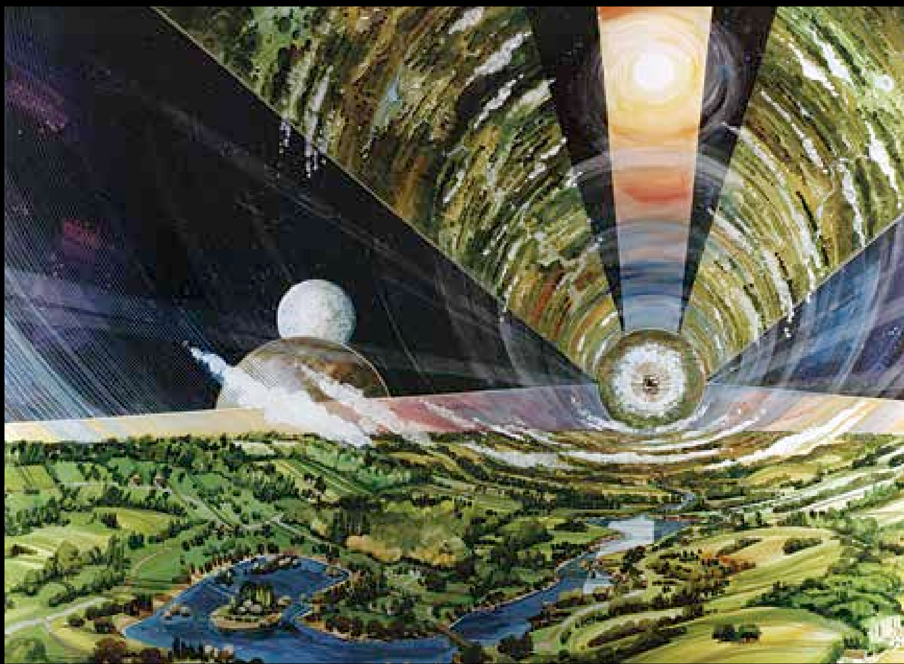
Imagine the year 2212 AD, and Interplanetary Star Cruiser (ISC) Antriksh gets ready to take off from its home spaceport on New Earth, a planet orbiting the triple star system, Alpha Centauri. This journey will be special, for at the other end lies the planet of origin of the human race, Earth. What sights will greet these descendents of Earth when they finally "come home"? How will they appear and function differently than we do right now? What social structures will bind them? How will they travel the long distance (25.4 trillion miles between the Sun and Alpha Centauri) that separates their origin and destination?

Isaac Asimov, Arthur C. Clarke and several others have long asked similar questions and come up with imagination-testing answers. But the threat of an apocalypse brought on either by environmental collapse, nuclear war, or an unsustainable population have made them increasingly relevant to our daily lives.



An illustration of the Rama spacecraft, which can be as long as 50 km in length

Setting up base

However, hope remains. Evidence of liquid water has been found on Jupiter's moon Europa and Saturn's moon Enceladus, making them prime candidates for a future human colony. Humans may also set up base on exoplanets (planets orbiting stars other than the sun), about 850 of which are already discovered, and the gradually increasing power of both Earthbound and space-based telescopes promise much more soon. The nearest of these, named Alpha Centauri Bb, orbits a triple star system 4.37 light years from the Sun.

Even this distance is too great for currently imaginable technology to cover in a single human lifespan. One possible solution would involve multiple generations of humans being born, living their lives, and dying in transit within the boundaries of the Solar system or in interstellar space. Spacecraft designed in the form of O'Neil cylinders (see the Rama spacecraft in "Rendezvous with Rama": http://youtu.be/zBIQCm54dfY) could support and shelter these humans in their long journey. These huge vehicles (50 km in length, 8 km in diameter) could produce artificial gravity at their inner surface by constantly rotating along their long axis, thus resulting in a centrifugal force that pushes everything away from the central axis and towards the inner curved surface. Furthermore, sufficient periods of day and night for the habitable areas would be achieved by opening and closing slits in sequence. Such a vehicle would serve the dual purpose of providing a controlled environment for habitation, agriculture, and industry, to sustain a large population and transport them across interstellar distances.

Adapting to space

But building an O'Neil cylinder would require large leaps in technology. We only have rocket-propelled spacecraft at our disposal currently, and using them for long-range travel poses several unique challenges. The lack of gravity over extended durations would result in low bone density and cause muscle atrophy. Our bodies would progressively weaken, and we would have to rely on personal robotic assistants for carrying us around and performing menial tasks, as glimpsed in the movie WALL-E. If we decided to rotate our spacecraft to artificially create gravity, we'd find it difficult to maintain a constant communication link with our home base. Once we wander beyond the moon's orbit, we would no longer be protected by the